Application of Sensors in Building Energy Activity Monitoring

Amber Sun, Anthony Darensbourg, Crystal Huang,

Hyungkyu Lee, Jason Hao, Leyna Dinh

Advisors Dr. Melody Baglione and Dr. Aaron Langham

ME-ECE393 Senior Capstone Design

Introduction

Of Cooper Union's three main buildings, 41 Cooper Square (41CS), uses a significantly larger portion of energy for its daily operation. In a 2020 study done by Cooper alumnus Nick Triano (ME '20), 41CS was found responsible for approximately 63% of Cooper's overall energy use [1]. With New York City's Local Law 97 (LL97) requiring greenhouse gas emissions limits on buildings over 25,000 square feet, 41CS needs to take actionable steps to reduce their emissions by 40% by 2030 and achieve net-zero building emissions by 2050 [2]. With current projections, however, 41CS stands to exceed the carbon emissions limit by 2035, which would incur an $87,661 yearly fee towards the school. Many student researchers have attempted to find ways to reduce emissions by tracking energy usage through predictive monitoring systems in the buildings.

a. Problem Statement:

Due to the variability in 41CS's mechanical operations and multiple power streams for the building's energy use, reporting accurate and consistent energy use proves to be a long-enduring task. Cooper Union facilities staff need a reliable way to understand how campus buildings are using energy because significant portions of the existing monitoring infrastructure are outdated, disconnected, or nonfunctional. As a result, equipment faults and energy waste can remain undetectable, increasing energy consumption and limiting informed facility decisions. Success means reliable, actionable visibility into building performance that helps staff identify faults, inefficient operation, and unnecessary energy use.

b. User/Application:

The target users for the high-frequency sensor and energy use dashboard are building managers who oversee the daily operations and consultants/researchers who are involved with retro-commissioning. Retro-commissioning is a common process where the existing building systems are evaluated to increase its performance and energy efficiency, overall helping improve occupant comfort. To understand what stakeholders would want out of a solution, the team spoke with John Rundell, a building consultant who works with Cooper Union facilities. He suggested

creating an energy dashboard that monitors every major point of inputs and outputs. Hard-wire inputs ranging from temperature sensors to flow meters would ideally display hourly data, a time scale useful for identifying energy efficiency. Outputs such as valve positions and fan pump speeds would help build a sense of fault detection. Another stakeholder would be Anatoly Lednyak, CEO of ALED Technologies, who works with sensors and data collection to support building energy monitoring for clients. He valued a bypass approach that involved additional sensors to gather data and upload to cloud for further analysis. However, he acknowledged that data collection should be filtered and gathered carefully to not overwhelm storage memory yet also provide sufficient information to make energy conclusions.

Background Research

Over the past decade, several Cooper alumni have attempted to tackle this issue. Nick Triano started by trying to define Cooper's energy usage. He used data points from 2018-2019 of building electricity and natural gas consumption, combined with weather patterns, to train a machine learning model to predict 41CS's electricity consumption well in advance [1]. Erin Ringold used a year's worth of monthly utility bills, with building occupancy rates, to develop a heat balance energy model for the 41CS building [3]. Oliver Zhang consolidated the building's data streams between the chillers, cooling towers, and pumps into one source for analysis using OSI PI. He used this information to develop a condenser water reset model [3]. Jon Rodriguez analyzed how efficient a cogeneration system would be in a building like 41 Cooper Square using Simulink [5].

Currently there are some systems in place to monitor energy usage throughout 41CS, but each of these systems have their own accompanying issues that restrict the accuracy of the acquired data. The Siemens Insight and Hubgrade Emsys systems used for live energy monitoring suffer from reduced data points from a combination of aging hardware and low frequency sampling rates. The Energy Star Portfolio and ConEdison energy usage reports that the Cooper Union only receives record data across 41CS as a single data point, which does not allow for data extrapolation on individual floor or device efficiency. The Cooper Union also used to have other energy monitoring dashboards such as Zhang's Cooper Power Dashboard and AVEVA API in the past. However, they are currently defunct due to broken hardware or

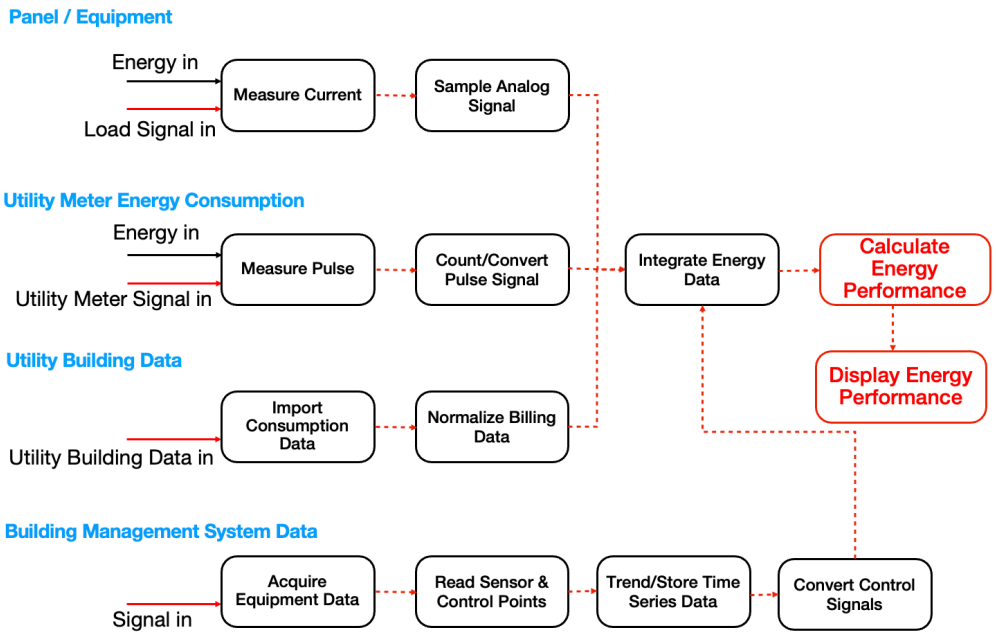
unsupported software. With the current data streams unable to provide consistent, accurate, or sufficient data points, a new model cannot be made for accurate predictions.

As mentioned earlier, interviews with John Rundell and Anatoly Lednyak revealed that an energy dashboard with consistent, reliable data streams and additional sensors would aid in more robust energy monitoring of 41CS. To aid us in achieving these goals at this point in the project, we decided to focus on investigating more into the cooling towers at the Cooper Union and current transformers (CTs) for use in additional sensors.

41CS has two cooling towers located on the roof of the building that are an integral part of the building temperature regulation system. Heated water that has been used in the evaporator of the chillers in the Lower Level 1 (LL1) floor gets sent to the cooling tower to release the stored heat into the ambient environment around Cooper. By running water down the open face of the cooling towers, some of the hot water evaporates into the air, releasing its energy and consequently cooling down the remaining water in the system. This cooled water is recollected at the bottom of the tower, sent to a water treatment tank since it was contaminated through exterior exposure, and returned to the chiller. The chiller systems in particular are an area of concern due to the many faulty sensors and the lack of consistent monitoring under the current systems despite its average excessive power consumption.

CTs, both the ones already installed in LL1 and the proposed additional hardware for high frequency monitoring of the elevators, function as a large transformer being spliced onto the power spine of the building. The voltage line being used to power sections of the building is fed through a magnetic metallic ring, representing one end of the transformer with only one loop. Then another wire is wrapped several times around the ring with a small resistor. A voltage reader is attached to the resistor to measure the voltage difference, and consequently the current running through the resistor. Combined with the known loop ratio of the transformer allows for easy calculation of the voltage and current running through the main power spine at any time. CTs become very useful in this instance due to their ability to step down the high currents on transmission lines to lower currents usable by voltage sensors and other circuit components.

Functional Decomposition



The functional decomposition above shows how energy and signals from various sensors flow through existing infrastructure and a potential solution. 41CS currently has two electrical meters, two natural gas meters, and two water meters that record usage in pulses of a fixed unit size, which is reflected in the utility meter branch of the diagram. The pulses from each meter have to be counted and converted to a usable signal before it can be used for energy monitoring. The Cooper Union also gets energy reports/bills from ConEdison and EnergyStar Portfolio for the building as a whole. The billing data and energy usage shown in the utility building data branch can be imported and processed to provide another signal for use downstream. 41CS also has building management systems like Desigo to monitor the status of building units such as chillers and heaters. The data from this equipment can be read and trended to produce data streams of control signals. These signals would then have to be converted into energy usage statistics for energy monitoring purposes.

On the panel/equipment branch of the inputs, our project would have to measure the current from equipment used in Cooper like the elevators or to monitor its power usage. Once this current is measured, the data must be sampled so that it can be stored for later use. With the inputs above, all this data needs to be aggregated into one location as shown in the “integrate energy data” block in order to evaluate the overall energy performance of the building. Other

metrics would need to be created and calculated on a dedicated display for monitoring use, represented by the red blocks in the diagram.

Technical Requirements

a. Functional Requirements

Block	Functional requirement
Measure Current	Shall measure RMS current for each monitored circuit and report invalid or out-of-range readings.
Measure Pulses	Shall detect and count meter pulses without missed or duplicate events.
Import Consumption Data	Shall import utility consumption data from CSV or API with timestamps, units, and meter ID.
Acquire Equipment Data	Shall acquire configured equipment and BMS points with value, timestamp, and quality status.
Sample Analog Signal	Shall sample analog signals faster than existing sensors and filter measurement noise.
Count/Convert Pulse Signal	Shall convert pulses using a configurable meter factor into interval and cumulative energy.
Normalize Billing Data	Shall normalize timestamps, units, intervals, and meter IDs and flag gaps or duplicates.
Read Sensor/Control Points	Shall read sensor values, setpoints, commands, and alarms with units and quality status.
Trend/Store Time Series Data	Shall store timestamped values and generate trends for selectable points and time ranges.
Convert Control Signals	Shall convert control values into standard engineering units and normalized equipment states.
Integrate Energy Data	Shall combine current, pulse, utility, and BMS data by timestamp, meter, and equipment ID.
Calculate Energy Performance	Shall calculate energy use, demand, cost, and performance metrics for selectable intervals.
Display Energy Performance	Shall display current and historical performance, equipment status, trends, and exceptions with configurable refresh intervals; point data shall support verification of configured ASHRAE control sequences.

b. Design Requirements and other Constraints

Block	Constraint (Metric+value)	Reason	Code/Standard
Measure Current	Install sensors on the transformer low-voltage side; measurement error ≤ ±2% over the operating range.	Reduces shock exposure while providing reliable circuit-level measurements.	NFPA 70; NFPA 70E; UL 61010-1

Display Energy Performance	Refresh live status within 60 s and show energy, demand, equipment state, trends, and exceptions.	Supports timely review of equipment activity and energy use.	ASHRAE Guideline 36-2024
Acquire / Integrate Data	Collect ≥95% of scheduled samples and align source timestamps within 60 s.	Maintains a complete, comparable record across meters, utility data, and BMS points.	ASHRAE 135 (BACnet)
Trend / Store Data	Store 15-minute interval data for at least 12 months; flag gaps and duplicate records.	Enables seasonal baselines, diagnostics, and performance verification.	ASHRAE Guideline 14
Convert Control Signals	Use read-only access by default; require authorization for commands and fail safely after communication loss.	Prevents unintended equipment operation and limits cybersecurity risk.	NIST SP 800-82; ASHRAE 135

c. Design Criteria

Category	Design Criterion	Target / Requirement	Method of Verification
Data	Data coverage	Automatically collect and store historical and live building-energy data with at least 95% availability during normal operation.	Compare expected and received records over a representative test period and calculate availability.
Data	Data quality & timestamp alignment	Use consistent units and equipment IDs. Flag missing or duplicate records; align source timestamps within 60 seconds where source systems permit.	Audit representative records and inject known gaps or duplicates; compare timestamps against a common reference.
Dashboard	Update latency	Automatically refresh new data and load typical dashboard plots within approximately 5 seconds at the expected pilot data volume.	Measure refresh and plot-loading times for a defined set of common dashboard tasks.
Dashboard	Usability	Display current use, historical trends, predicted load, and abnormal conditions without requiring users to analyze raw data.	Conduct task-based review with Facilities personnel, researchers, or representative project users.
Machine Learning	Load-prediction accuracy	Evaluate MAE and RMSE and, where valid, MAPE; use an initial target below approximately 15% MAPE and revise it based on available data quality.	Evaluate the model on held-out historical data and document metrics, assumptions, and any revised threshold.
High-Frequency CT	Elevator event capture	Use a minimum sampling frequency established through testing, beginning in the hundreds of samples per second or higher, sufficient to distinguish startup, travel, stopping, and idle events.	Record repeated elevator trips at candidate sampling rates and compare preservation and repeatability of event features.

Anomaly Detection	Anomaly identification	Create a normal-operation baseline and produce a timestamped anomaly score or classification for each elevator event; do not claim a specific fault without experimental verification.	Review event outputs against labeled normal and available simulated or abnormal data; report precision, recall, detection rate, and false-positive rate when sufficient labels exist.
System Integration	Integration effort	Use standardized timestamps, equipment names, and documented interfaces; failure of the CT subsystem must not stop the building-energy data stream or dashboard.	Perform an end-to-end integration test and a controlled CT-subsystem shutdown while verifying continued dashboard operation.
System Architecture	Scalability & maintainability	Support the planned nine CT channels and a path to at least 12 channels without core redesign; document configuration, data paths, code, and component interfaces.	Add simulated or spare channels and have another team member follow the documentation to configure or restart a component.
Safety & Security	Safe, read-only monitoring	Use non-invasive sensing where feasible; do not command or alter elevator controls; electrical-panel work must be performed or supervised by qualified personnel.	Review wiring, permissions, installation plans, and safety approvals; verify normal elevator operation before and after installation.

d. Solution comparison

For the scoring of each criterion, each score ranges from –2 to 2 where positive scores represent better performance and negative scores represent worse performance.

Criterion	**Concept 1 Existing BMS + manual import**	**Concept 2 Edge gateway + local dashboard**	**Concept 3 Cloud analytics platform**
Data coverage	0	+2	+2
Data quality & timestamp alignment	-1	+2	+2
Update latency	0	+2	+1
Usability	-1	+2	+2
Load-prediction accuracy	-1	+1	+2
Integration effort	0	-1	-2

Criterion	Concept 1 Existing BMS + manual import	Concept 2 Edge gateway + local dashboard	Concept 3 Cloud analytics platform
Scalability & maintainability	0	+1	+2
Net	**-3**	**+9**	**+9**

High-Frequency Fault Detection Team

Criterion	Concept 1 Existing elevator / BMS data	Concept 2 Centralized high-frequency DAQ	Concept 3 CT + Raspberry Pi edge processing
Elevator event capture	-2	+1	+2
Anomaly identification	-2	+1	+2
Integration effort	+2	-1	0
Scalability & maintainability	0	-1	+2
Safe, read-only monitoring	+1	+1	+2
Net	**-1**	**+1**	**+8**

e. Safety Review

Block	Reason
Measure Current	Voltages above 120Vrms, currents above 200mA, or a capacitor or inductor storing more than 5 Joules
Acquire/Integrate Data	Gross physical hazards to humans or the environment, such as experiments using the roof or in mechanical equipment rooms (MER)

Please refer to the project safety form located in Appendix I.

Proposed Work

Because the project team is relatively large and the project contains two technically distinct workstreams, the team has been divided the proposed work into two sub teams: the Legacy Data Team, focused on restoring the historical energy data pipeline, dashboard, and load-prediction model, and the High-Frequency Fault Detection Team, focused on CT-based elevator monitoring and anomaly detection.

a. Legacy Data Team

41CS's previous monitoring systems are mostly operative with specific sections having either damaged, deprecated, or removed components. Rather than fully replacing a semi-functioning design, those components are being replaced or updated to revitalize the system. For hardware that has outright broken down or represented sources of data leaks– such as the SATEC meter in LL1– replacements were ordered, and installation has been completed. The next step of the process is to rebuild the user dashboard; highlighting an updated predictive model and a more intuitive design of the building's energy data. This intuitive design will include data previously unrepresented and display data in an intuitive manner for inexperienced users. Previous dashboards will be used to inform the decision-making process. Meetings will be conducted to collect constructive feedback on the design from stakeholders and advisors along the way. Lastly, a predictive model will be developed in the dashboard based on input data from the current low-frequency active monitoring systems to estimate building energy use and the resulting costs.

b. High-Frequency Fault Detection Team

To monitor the energy use of the three 41CS elevators, our prototype will take the form of a power monitoring box. The current from the elevator transmission lines will be measured with CTs rated at 100A that will be installed along the voltage inputs for the elevator motors. The outputs of the CTs are connected to a voltage sensor/Raspberry Pi circuit that will sample and process the CT data into recognizable current waves and send it to a centralized data acquisition point for interpretation.

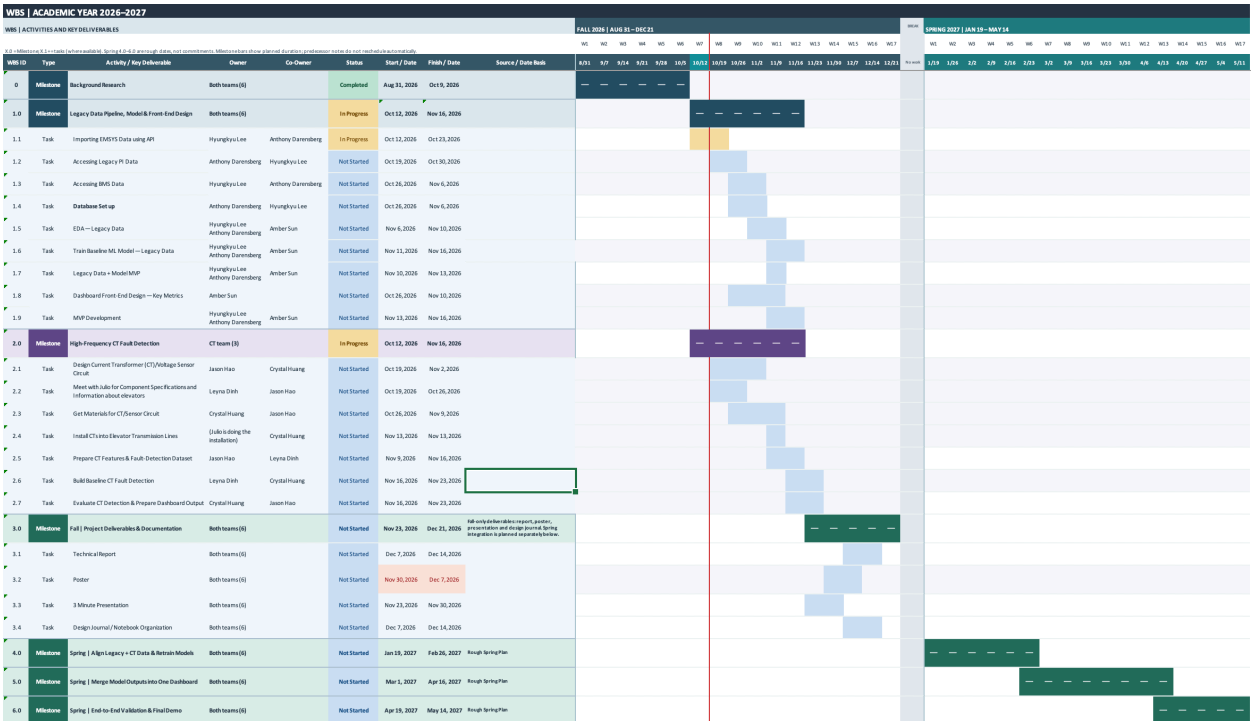
The biggest obstacles that will most likely be encountered are the installation process and the placement of the CTs in the elevator lines. Currently, it is unclear whether placing the CTs before or after the drive box powering the elevator motors would provide us with better data. In addition, the wiring of the elevator motors would need to be disconnected to install the CTs, which would disrupt elevator service in the meantime. The current plan is to organize a time with the Facilities staff to first install the CT/sensor circuit with their help for just one elevator to test

the efficacy of the CT placement. This feedback can then be used to change the design of the CT circuits for the other two elevators in 41CS.

A bill of materials for the components that are expected to be used in this prototype is given in Appendix II of the document. Most of the components will be used for creating and installing the CT circuit into the elevator, but a new Satec meter is needed to replace the broken Satec meter in LL1. As of now, the new Satec meter has already been purchased and installed in the switch room on September 30th, 2026.

Please refer to the bill of materials located in Appendix II.

Timeline and Project Management



WBS | ACADEMIC YEAR 2026–2027

WBS | ACTIVITIES AND KEY DELIVERABLES

WBS ID	Type	Activity / Key Deliverable	Owner	Co-Owner	Status	Start / Date	Finish / Date	Source / Date Basis
0	Milestone	Background Research	Both teams (6)		Completed	Aug 31, 2026	Oct 9, 2026	
1.0	Milestone	Legacy Data Pipeline, Model & Front-End Design	Both teams (6)		In Progress	Oct 12, 2026	Nov 16, 2026	
1.1	Task	Importing EMSYS Data using API	Hyungkyu Lee	Anthony Darensberg	In Progress	Oct 12, 2026	Oct 23, 2026	
1.2	Task	Accessing Legacy PI Data	Anthony Darensberg	Hyungkyu Lee	Not Started	Oct 19, 2026	Oct 30, 2026	
1.3	Task	Accessing BMS Data	Hyungkyu Lee	Anthony Darensberg	Not Started	Oct 26, 2026	Nov 6, 2026	
1.4	Task	Database Set up	Anthony Darensberg	Hyungkyu Lee	Not Started	Oct 26, 2026	Nov 6, 2026	
1.5	Task	EDA — Legacy Data	Hyungkyu Lee Anthony Darensberg	Amber Sun	Not Started	Nov 6, 2026	Nov 10, 2026	
1.6	Task	Train Baseline ML Model — Legacy Data	Hyungkyu Lee Anthony Darensberg	Amber Sun	Not Started	Nov 11, 2026	Nov 16, 2026	
1.7	Task	Legacy Data + Model MVP	Hyungkyu Lee Anthony Darensberg	Amber Sun	Not Started	Nov 10, 2026	Nov 13, 2026	
1.8	Task	Dashboard Front-End Design — Key Metrics	Amber Sun		Not Started	Oct 26, 2026	Nov 10, 2026	
1.9	Task	MVP Development	Hyungkyu Lee Anthony Darensberg	Amber Sun	Not Started	Nov 13, 2026	Nov 16, 2026	
2.0	Milestone	High-Frequency CT Fault Detection	CT team (3)		In Progress	Oct 12, 2026	Nov 16, 2026	
2.1	Task	Design Current Transformer (CT)/Voltage Sensor Circuit	Jason Hao	Crystal Huang	Not Started	Oct 19, 2026	Nov 2, 2026	
2.2	Task	Meet with Julio for Component Specifications and Information about elevators	Leyna Dinh	Jason Hao	Not Started	Oct 19, 2026	Oct 26, 2026	
2.3	Task	Get Materials for CT/Sensor Circuit	Crystal Huang	Jason Hao	Not Started	Oct 26, 2026	Nov 9, 2026	
2.4	Task	Install CTs into Elevator Transmission Lines	(Julio is doing the installation)	Crystal Huang	Not Started	Nov 13, 2026	Nov 13, 2026	
2.5	Task	Prepare CT Features & Fault-Detection Dataset	Jason Hao	Leyna Dinh	Not Started	Nov 9, 2026	Nov 16, 2026	
2.6	Task	Build Baseline CT Fault Detection	Leyna Dinh	Crystal Huang	Not Started	Nov 16, 2026	Nov 23, 2026	
2.7	Task	Evaluate CT Detection & Prepare Dashboard Output	Crystal Huang	Jason Hao	Not Started	Nov 16, 2026	Nov 23, 2026	
3.0	Milestone	Fall | Project Deliverables & Documentation	Both teams (6)		Not Started	Nov 23, 2026	Dec 21, 2026	fall-only deliverables: report, poster, presentation and design journal. Spring integration is planned separately below.
3.1	Task	Technical Report	Both teams (6)		Not Started	Dec 7, 2026	Dec 14, 2026	
3.2	Task	Poster	Both teams (6)		Not Started	Nov 30, 2026	Dec 7, 2026	
3.3	Task	3 Minute Presentation	Both teams (6)		Not Started	Nov 23, 2026	Nov 30, 2026	
3.4	Task	Design Journal / Notebook Organization	Both teams (6)		Not Started	Dec 7, 2026	Dec 14, 2026	
4.0	Milestone	Spring | Align Legacy + CT Data & Retrain Models	Both teams (6)		Not Started	Jan 19, 2027	Feb 26, 2027	Rough Spring Plan
5.0	Milestone	Spring | Merge Model Outputs into One Dashboard	Both teams (6)		Not Started	Mar 1, 2027	Apr 16, 2027	Rough Spring Plan
6.0	Milestone	Spring | End-to-End Validation & Final Demo	Both teams (6)		Not Started	Apr 19, 2027	May 14, 2027	Rough Spring Plan

In order to track the tasks that need to be completed for each subteam, we have created a Gantt chart to display who is assigned to each task and the rough timeframe to work on each task. A link to the Gantt chart above can be found here. Each task on the project timeline is briefly explained in more detail below.

- **1. Historical Data Pipeline / Dashboard Restoration**
    - 1.1 Importing Emsys Data Using API: Develop and test the API connection needed to retrieve current energy data from Emsys.

- 1.2 Accessing Legacy PI Data: Recover and verify access to historical building energy data stored in Cooper's previous PI system.
- 1.3 Accessing BMS Data: Create a data pipeline to fetch BMS data.
- 1.4 Database Setup: Set up the database structure needed to store, organize, and retrieve restored historical and incoming energy data.
- 1.5 EDA – Legacy Data: Perform exploratory data analysis on the historical dataset to understand trends, missing values, anomalies, and overall data quality.
- 1.6 Train Baseline ML Model – Develop an initial machine-learning model to predict building energy consumption based on historical energy data and weather.
- 1.7 Legacy Data + Model MVP: Create a dashboard-style system that combines the historical dataset and prediction model to compare actual and predicted energy behavior.
- 1.8 Dashboard Front-End Design – Key Metrics: Design the dashboard interface and determine which energy metrics, trends, and prediction results should be displayed to users.
- 1.9 MVP Development: Integrate the data pipeline, dashboard, and machine-learning components into a functional prototype.

- **2. High-Frequency CT Fault Detection**
  - 2.1 Design Current Transformer / Voltage Sensor Circuit: Design the sensing circuit that will safely measure high-frequency electrical current and, if required, voltage from the elevator system.
  - 2.2 Meet with Julio to Compare Specifications and Information About Elevator: Coordinate with Facilities to understand the elevator's electrical characteristics, operating constraints, and suitable monitoring locations.
  - 2.3 Get Hardware / CT Sensor Circuit: Prepare the CT sensors, data-acquisition hardware, and supporting electrical components.
  - 2.4 Install CT into Elevator Transmission Lines: Install the sensing hardware on elevator electrical lines without interfering with normal elevator operation.
  - 2.5 Prepare CT Features and Fault-Detection Dataset: Process the high-frequency CT measurements and extract useful signal features for distinguishing normal and potentially abnormal elevator behavior.

    - 2.6 Build Baseline CT Fault-Detection: Develop an initial anomaly-detection model using the elevator CT data to establish a baseline of normal operation.
    - 2.7 Evaluate CT Detection & Prepare Dashboard Output: Assess the anomaly-detection results and determine how detected events, anomaly scores, and CT signals should be presented on the dashboard.
- **3. Project Deliverables & Documentation**
    - 3.1 Technical Report: Formal step by step explanation of work completed for the project
    - 3.2 Poster: Create a poster explaining the problem statement and show casing the research and work done for the project throughout the semester
    - 3.3 3 Minute Presentation: Elevator pitch style explanation of work done for this project
    - 3.4 Design Notebook Organization: Organize notes taken over the course of the semester from meetings, research and work done as a legacy of the project
- **4. Spring - Align Legacy + CT Data & Retrain Models:** Integrate the restored building-energy data with the high-frequency CT monitoring system, refine the overall architecture, and retrain the prediction and anomaly-detection models using additional data.
- **5. Spring - Merge Model Outputs into One Dashboard:** Combine building load predictions, historical/current energy data, and elevator anomaly detection results in a single unified monitoring dashboard.
- **6. Spring - End-to-End Validation & Final Demo:** Test the complete system from data acquisition through analysis and visualization, validate that the components work together reliably, and prepare the final project demonstration.

Bibliography

[1] N. Triano, “Data Analysis Tools for Evaluating Building Energy Consumption and Carbon Emissions Goals,” presented to ME499, The Cooper Union, New York, New York, USA [PowerPoint slides].

This source, created by Cooper alumnus Nick Triano for a master’s thesis defense, provides an insight into the energy usage of 41CS. The author produces a machine learning model to predict hourly electricity consumption based on weather features (such as relative humidity and air temperature) and temporal features (such as day or the week or hour). Additionally, the author improves the resolution of incoming natural gas data, while also creating a web application dashboard to calculate Cooper’s carbon emissions, energy costs, and energy use. The author highlights the need for increasing dependence on electricity over natural gas, in order to meet NYC’s Local Law 97 target and reduce carbon emissions.

[2] New York City Department of Buildings, “LL97 Greenhouse Gas Emissions Reduction,” *NYC.gov*, New York, NY, USA. Accessed: Sep. 29, 2026.

Local Law 97 of 2019 (LL97) is a piece of New York City legislation that is intended to curtail carbon emissions in the city by 2050. It institutes limits on carbon emissions for large buildings and institutes monetary penalties for buildings that do not abide by the limits.

Since all three of the buildings on the Cooper Union’s campus are affected by the carbon emissions restriction outlined by LL97, this law acts as an impetus for projects like ours to try and find ways to monitor and reduce energy consumption on campus. Considering that all of the buildings are projected to be above the limits set by 2035, it also gives a measurable cost to the school if no action was taken towards improving emissions.

[3] E. Ringold, “Development of a Calibrated Whole-Building Heat Balance Energy Model for a High-Performance Academic Building (41 Cooper Square, NY, NY),” The Cooper Union, New York, New York, USA, Sept. 14, 2016 [PowerPoint slides].

Alumnus Eric Ringold’s master’s thesis very closely resembles the scope of the current project. The author’s research focuses on modeling 41CS and simulating the energy consumption of the building using EnergyPlus, a Building Energy Modeling (BEM) software engine using physics-based calculations. After modeling surface material, occupancy zones, and HVAC systems, the author calibrated the simulation using building energy data sourced from utility bills and cogeneration system usage.

[4] O. Zhang, “Development of Building Data Management Software and Analysis of Energy Savings from a Condenser Water Reset Strategy ,” The Cooper Union, New York, New York, USA [PowerPoint slides].

Oliver Zheng, another Cooper alumnus, consolidated building energy data using OSI PI, a data management software. The author created a platform to more effectively display energy information to students, faculty, and building staff. A simulation of energy savings from a theoretical chiller plant Condenser Water Reset strategy was created, aiming to decrease carbon emissions and reduce fines related to LL97.

[5] J. Rodriguez, “Technical and Economic Assessment of a Cogeneration System in an Urban Academic Building ,” The Cooper Union, New York, New York, USA, Apr. 28, 2014 [PowerPoint slides].

Alumnus Jonathan Rodriguez’s master’s thesis describes the workings and benefits of a cogeneration system, also known as Combined Heat and Power (CHP). In a CHP system, a heat engine turns fuel into both electricity and heat simultaneously, reducing utility costs. Natural gas is around 5 times cheaper than electricity, so reducing reliance on electricity straight from the provider like ConEdison will save money on operating costs. The author ran 3 simulations of 41CS in Simulink to determine the effects and limitations of the existing 41CS CHP system, and proposed a retrofit solution of a properly sized heat exchanger to reduce demand on the boilers while increasing savings. The results indicated savings of $206k with the retrofitted CHP system.

Appendix I. Safety Review

The Cooper Union for the Advancement of Science and Art
41 Cooper Square
New York, NY 10003

Capstone Project Safety Review

Your capstone project faculty and the Cooper Union Safety Office want to ensure your capstone project is built and tested in a safe manner. As part of the capstone project design process, please fill out this safety review. If necessary, you may need to obtain the signature of a Campus-Wide Safety Coordinator to ensure that the safety protocols that you have established are consistent with acceptable practice.

Researcher(s): Amber Sun, Anthony Darensbourg, Crystal Huang, Leyna Dinh, Hyungkyu Lee, Jason Hao

Project Advisor: Professor Baglione and Professor Langham

Title of Project: Building Energy Activity Monitoring

Please place an "X" in all boxes below which apply to your capstone project:

[ ] Chemicals, including photographic fixers, etc., essentially anything other than water.
[ ] Drone testing
[X] Voltages above 120Vrms, currents above 200mA, or a capacitor or inductor storing more than 5 Joules
[ ] Lasers
[ ] Instantaneous or percussive audio signals exceeding 115dB
[X] Gross physical hazards to humans or the environment, such as experiments using the roof or in mechanical equipment rooms (MER)
[ ] Gas, including non-toxic high-pressure ones, such as air or steam
[ ] Power tools beyond the ones covered in EID 101 safety training
[ ] Soldering, or the generation of toxic smoke
[ ] Poisons, including radioactive sources, lead etc.
[ ] UV (ultraviolet) light
[ ] Glass or "sharps," such as needles
[ ] Bio-hazards such as bacteria or blood
[ ] The potential for misuse of the "buddy" system, single students unsupervised and alone in a lab or workshop
[ ] Projectile systems
[ ] Heat, ovens and the potential for burning
[ ] Generation of fumes and/or vapor, including laser cutters or non-PLA 3D printers

[ ] Activities which might cause "repetitive stress" type injuries (other than normal typing) or projects with components that attach to people (e.g. prosthetics or exo-skeletons).
[ ] Physical impact, such as running on a treadmill
[ ] Microwave or radio frequencies at power levels above 1W EIRP
[ ] Moving objects heavier than 50 lbs.
[ ] Use of heavy machinery such as material tensile strength (Instron) machines, large robot arms
[ ] Use of LiPo batteries
[ ] Other, self-identified hazard:
Please explain / identify:

If you have checked one or more of the boxes on the first page of this form, please answer the questions below and obtain the signatures of Associate Dean Ruben Savizky or Professor Jennifer Weiser, the Campus-Wide Safety Coordinators.

1. Which course is this work being conducted in?

Course number(s)_____ECE395/ME393_____

2. Who will be the subjects of the proposed research? If the subjects are people not on your capstone project team, also submit an IRB application. See https://cooper.edu/about/policies/human-subject-research

There are no subjects in the proposed research.

3. Who will carry out the proposed research?

Crystal Huang, Jason Hao, Leyna Dinh, Amber Sun, Hyungkyu Lee, and Anthony Darensbourg

4. Please provide a brief project description.

The project involves improving the energy monitoring systems at 41 Cooper Square. In particular, one aspect of the project involves monitoring the energy usage of the elevators since the current energy usage is sampled too slowly for use in fault detection.

To obtain higher frequency samples for the elevator, this project will involve installing current transformers to the elevator lines to monitor the current being used. This current data can then be used to calculate the power usage and fault detection in the 41CS elevators at a more granular level.

5. Please describe the specific hazards involved and how risks will be addressed and/or minimized during the research.

We intend to install current transformers in the elevator motor room which houses high current wires and motors. To mitigate the risks involved with the installation, we intend to coordinate with the Facilities staff to arrange a time for one of the staff to install the transformers while the elevator is put out of service.

_____ _____

Signature of Researcher (s) **Date**

_____ _____

Signature of Project Advisor(s) **Date**

_____ _____

Signature of Laboratory Technician(s) responsible for lab(s) where testing will occur **Date**

_____ _____

Signature of Campus-Wide Safety Coordinator **Date**

Appendix II. Bill of Materials

Item	Price	Amount	Source
100A Current Transformers	$70.00	9	AimDynamics
Raspberry PiHat	$99.00	1	DigiKey
Raspberry Pi 5 (4 GB)	$111.00	1	Waveshare
High Voltage Sensor	$194.25	3	eGauge
Replace Satec Meter	$5,000.00	1	Veolia Quote
Total:	$6,422.75	**(w/o Satec meter)**	$1422.75